NEET UG BIOLOGY • COMPLETE STUDY MODULE

# Cell Cycle and Cell Division

**Syllabus Context:** Cell Biology — Phases of Cell Cycle, Mitosis (Karyokinesis and Cytokinesis), Significance of Mitosis, Meiosis I and Meiosis II, and Significance of Meiosis.

## 1. Learning Objectives & Overview

All living organisms, even the largest multicellular species, begin their life from a single cell (such as a single-celled zygote in sexually reproducing organisms). Growth and reproduction are fundamental characteristics of cells and of all living organisms. Cells reproduce by dividing into two, where each parental cell gives rise to two daughter cells each time it divides. These newly formed daughter cells grow and divide further, forming a new cell population derived from a single parental cell and its progeny.

**After studying this tutorial, you will be able to:**

- Define the cell cycle and distinguish between Interphase ($\mathrm{G_1}$, $\mathrm{S}$, $\mathrm{G_2}$), the quiescent stage ($\mathrm{G_0}$), and the $\mathrm{M}$ phase.
- Track the exact changes in chromosome number ($n$ or $2n$) and DNA content ($C$ value) across all stages of mitosis and meiosis.
- Identify the characteristic cytological events of Prophase, Metaphase, Anaphase, and Telophase in mitosis, and contrast animal and plant cytokinesis.
- Analyze the five substages of Prophase I (Leptotene, Zygotene, Pachytene, Diplotene, and Diakinesis) and compare Meiosis I (reductional division) with Meiosis II (equational division).
- Solve high-yield NEET UG conceptual, numerical, statement-based, and assertion-reason questions with precision.

## 2. The Cell Cycle and Its Phases

During the division of a cell, DNA replication and cell growth must take place in a coordinated manner to ensure correct division and the formation of progeny cells containing intact genomes.

**Definition — Cell Cycle:** The sequence of events by which a cell duplicates its genome, synthesises the other constituents of the cell, and eventually divides into two daughter cells is termed the **cell cycle**.

Although cell growth (in terms of cytoplasmic increase) is a continuous process, DNA synthesis occurs only during one specific stage in the cell cycle. The replicated chromosomes (DNA) are then distributed to daughter nuclei by a complex series of events during cell division. All these events are under strict genetic control.

### 2.1 Duration of the Cell Cycle

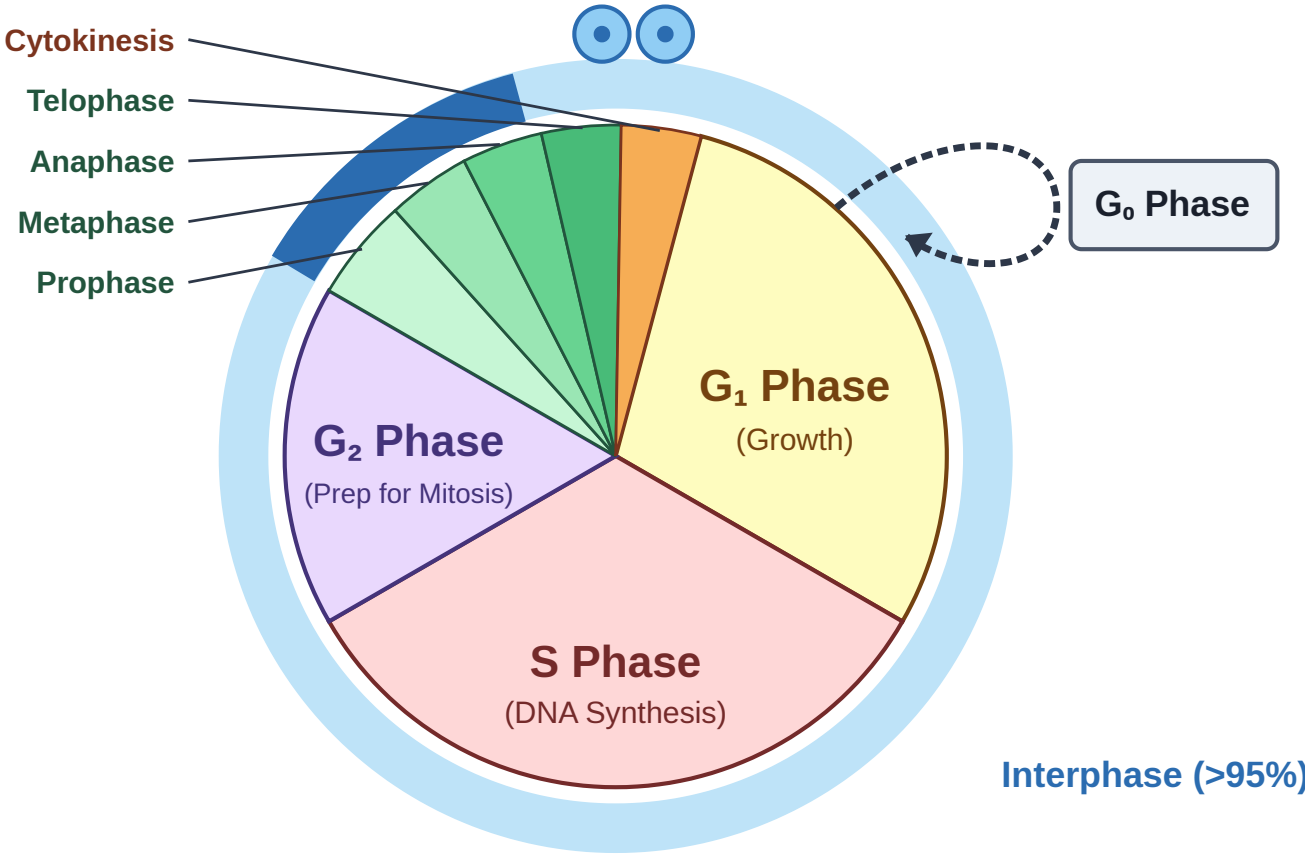
The duration of the cell cycle varies from organism to organism and from cell type to cell type:

- **Human cells in culture:** Divide once in approximately every $24\ \mathrm{hours}$. Of this $24\text{-hour}$ average duration, cell division proper ($\mathrm{M}$ phase) lasts for only about $1\ \mathrm{hour}$, while **Interphase lasts more than** $95\%$ of the duration of the cell cycle.
- **Yeast:** Can progress through the entire cell cycle in only about $90\ \mathrm{minutes}$.

### 2.2 Basic Phases of the Cell Cycle

A typical eukaryotic cell cycle is divided into two basic phases:

1. **Interphase:** Represents the phase between two successive $\mathrm{M}$ phases. Though traditionally called the *resting phase*, it is the time during which the cell prepares for division by undergoing both cell growth and DNA replication in an orderly manner.
2. $\mathrm{M}$ **Phase (Mitosis phase):** Represents the phase when actual cell division occurs. It starts with nuclear division, corresponding to the separation of daughter chromosomes (**karyokinesis**), and usually ends with the division of cytoplasm (**cytokinesis**).




**Tutorial Diagram 1: Diagrammatic view of the eukaryotic cell cycle showing Interphase ($G_1$, S, $G_2$), M Phase (Prophase, Metaphase, Anaphase, Telophase, and Cytokinesis), and the quiescent stage ($G_0$).**

### 2.3 Subphases of Interphase

Interphase is divided into three sequential phases:

- $\mathrm{G}_1$ **phase (Gap 1):** Corresponds to the interval between mitosis and the initiation of DNA replication. During $\mathrm{G}_1$ phase, the cell is metabolically active, carries out normal metabolism, and continuously grows, but **does not replicate its DNA**. Most organelle duplication also occurs during this phase.
- $\mathrm{S}$ **phase (Synthesis phase):** Marks the period during which **DNA synthesis or replication** and chromosome duplication take place.
  - During this time, the **amount of DNA per cell doubles**. If the initial amount of DNA is denoted as $2C$, then it increases to $4C$.
  - However, there is **no increase in the chromosome number**; if the cell had a diploid or $2n$ number of chromosomes at $\mathrm{G}_1$, even after the $\mathrm{S}$ phase the number of chromosomes remains the same, i.e., $2n$.
  - In animal cells, during the $\mathrm{S}$ phase, DNA replication begins in the **nucleus**, and the **centriole duplicates in the cytoplasm**.
- $\mathrm{G}_2$ **phase (Gap 2):** During the $\mathrm{G}_2$ phase, **proteins are synthesised** in preparation for mitosis while cytoplasmic cell growth continues.

### 2.4 The Quiescent Stage ($\mathrm{G}_0$) and Tissue Division Patterns

Not all cells in an adult organism divide continuously:

- Some cells in adult animals do not appear to exhibit division (e.g., **heart cells**), and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death.
- These cells that do not divide further exit the $\mathrm{G}_1$ phase to enter an inactive stage called the **quiescent stage ($\mathrm{G}_0$)** of the cell cycle.
- Cells in the $\mathrm{G}_0$ stage **remain metabolically active** but no longer proliferate unless called on to do so depending on the requirement of the organism.
- **Plants vs. Animals:** Higher plants continue to grow throughout their lives due to mitotic divisions in specialised **meristematic tissues**—specifically the **apical and lateral cambium**. Animals do not possess such meristematic tissues.
- **Ploidy of Mitotically Dividing Cells:** In animals, mitotic cell division is generally seen only in **diploid somatic cells**. However, there are a few exceptions where haploid cells divide by mitosis, for example, **male honey bees (drones)**. In contrast, plants

can show mitotic divisions in **both haploid and diploid cells** (representing the gametophytic and sporophytic generations, respectively, in plants exhibiting alternation of generations).

## 3. M Phase (Mitosis)

The $\mathrm{M}$ phase is the most dramatic period of the cell cycle, involving a major reorganisation of virtually all components of the cell. Since the number of chromosomes in the parent and progeny cells is conserved (remains the same), mitosis is called **equational division**.

Although mitosis is divided into four stages of nuclear division (**karyokinesis**) for convenience, cell division is a progressive process and very clear-cut lines cannot be drawn between various stages. Karyokinesis involves four sequential stages:

1. **Prophase**
2. **Metaphase**
3. **Anaphase**
4. **Telophase**

### 3.1 Prophase

Prophase is the first stage of karyokinesis of mitosis and follows the $\mathrm{S}$ and $\mathrm{G}_2$ phases of interphase. In the $\mathrm{S}$ and $\mathrm{G}_2$ phases, the new DNA molecules formed are not distinct but intertwined. Prophase is marked by the **initiation of condensation of chromosomal material**, during which the chromatin becomes untangled.

**Key characteristic events marking the completion of prophase:**

- Chromosomal material condenses to form compact mitotic chromosomes. Chromosomes are seen to be composed of **two chromatids attached together at the centromere**.
- The **centrosome**, which had undergone duplication during the $\mathrm{S}$ phase of interphase, begins to move towards opposite poles of the cell.
- Each centrosome radiates out microtubules called **asters**. The **two asters together with spindle fibres form the mitotic apparatus**.
- Cells at the end of prophase, when viewed under the microscope, **do not show Golgi complexes, endoplasmic reticulum, nucleolus, and the nuclear envelope**.

### 3.2 Metaphase

The **complete disintegration of the nuclear envelope** marks the start of the second phase of mitosis (metaphase); consequently, the chromosomes are spread through the cytoplasm of the cell. By this stage, condensation of chromosomes is completed and they can be observed clearly under the microscope. Therefore, **metaphase is the stage at which the morphology of chromosomes is most easily studied**.

A metaphase chromosome is made up of two sister chromatids held together by the centromere. Small disc-shaped structures at the surface of the centromeres are called **kinetochores**. Kinetochores serve as the sites of attachment of spindle fibres to the chromosomes that are moved into position at the centre of the cell. The plane of alignment of the chromosomes at metaphase is referred to as the **metaphase plate**.

**Key features of metaphase:**

- Spindle fibres attach to the **kinetochores** of chromosomes.
- Chromosomes are moved to the spindle equator and get aligned along the **metaphase plate** through spindle fibres connected to both poles (one chromatid of each chromosome is connected by its kinetochore to spindle fibres from one pole, and its sister chromatid is connected by its kinetochore to spindle fibres from the opposite pole).

### 3.3 Anaphase

At the onset of anaphase, each chromosome arranged at the metaphase plate is split simultaneously, and the two daughter chromatids—now referred to as **daughter chromosomes of the future daughter nuclei**—begin their migration towards the two opposite poles. As each chromosome moves away from the equatorial plate, the **centromere of each chromosome remains directed towards the pole and hence is at the leading edge, with the arms of the chromosome trailing behind**.

**Key events characterizing anaphase:**

- **Centromeres split** and chromatids separate.
- **Chromatids (daughter chromosomes) move to opposite poles.**

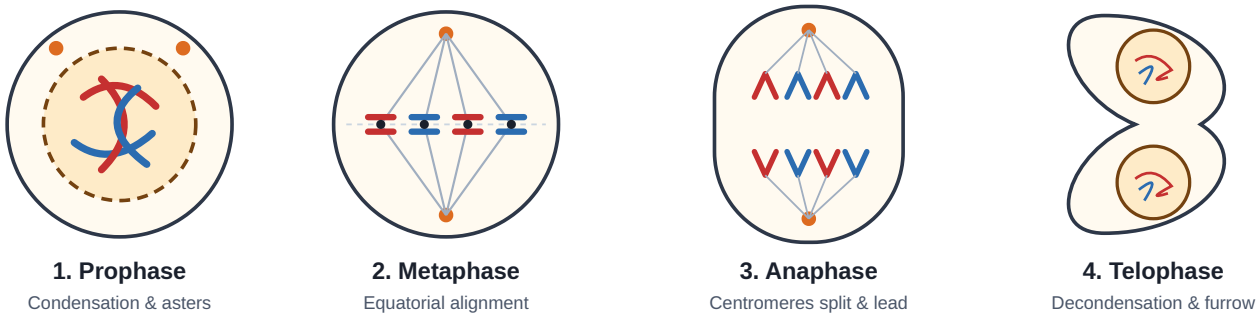
### 3.4 Telophase

At the beginning of the final stage of karyokinesis (telophase), the chromosomes that have reached their respective poles decondense and lose their individuality. Individual chromosomes can no longer be seen, and each set of chromatin material tends to collect at each of the two poles.

**Key events of telophase:**

- Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.
- **Nuclear envelope develops** around the chromosome clusters at each pole, forming two daughter nuclei.

- **Nucleolus, Golgi complex, and endoplasmic reticulum (ER) reform.**




**Tutorial Diagram 2: Stages of Mitosis (2n = 4 illustrated) showing chromosome condensation in Prophase, equatorial alignment in Metaphase, centromere splitting with centromeres at the leading edge in Anaphase, and nuclear reformation in Telophase.**

### 3.5 Cytokinesis

Mitosis accomplishes not only the segregation of duplicated chromosomes into daughter nuclei (karyokinesis), but the cell itself is divided into two daughter cells by the separation of cytoplasm called **cytokinesis**, at the end of which cell division gets completed. At the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between the two daughter cells.

| Feature | Animal Cell Cytokinesis | Plant Cell Cytokinesis |
|---|---|---|
| **Outer Boundary Constraint** | Flexible plasma membrane without a cell wall. | Enclosed by a relatively inextensible, rigid cell wall. |
| **Mechanism of Division** | Achieved by the appearance of a **furrow** in the plasma membrane. | Achieved by the formation of a new cell wall precursor called the **cell-plate**. |
| **Direction of Progression** | **Centripetal**: The furrow starts at the periphery, gradually deepens, and ultimately joins in the centre dividing the cytoplasm into two. | **Centrifugal**: Wall formation starts in the centre of the cell and grows outward to meet the existing lateral walls. |
| **Structural Fate** | Pinches the parent cell membrane directly into two daughter cells. | The **cell-plate represents the middle lamella** between the walls of two adjacent cells. |

**Syncytium Formation:** In some organisms, karyokinesis is **not followed by cytokinesis**, as a result of which a multinucleate condition arises leading to the formation of a **syncytium** (e.g., **liquid endosperm in coconut**).

### 3.6 Significance of Mitosis

- **Genetic Stability:** Mitosis usually results in the production of diploid daughter cells with an **identical genetic complement**.
- **Growth of Multicellular Organisms:** The growth of multicellular organisms from a single-celled zygote is due to mitosis.
- **Restoration of Nucleo-Cytoplasmic Ratio:** Continuous cell growth disturbs the ratio between the nucleus and the cytoplasm. Mitotic division is essential to restore the proper **nucleo-cytoplasmic ratio**.
- **Cell Repair and Replacement:** A very significant contribution of mitosis is cell repair. The cells of the **upper layer of the epidermis**, cells of the **lining of the gut**, and **blood cells** are constantly replaced by mitosis.
- **Continuous Plant Growth:** Mitotic divisions in meristematic tissues (apical and lateral cambium) enable continuous growth of plants throughout their life.
- **Haploid Mitosis in Special Cases:** While mitosis is usually restricted to diploid cells, in some lower plants and in some social insects (such as male honey bees), haploid cells also divide by mitosis.

## 4. Meiosis (Reductional Division)

Sexual reproduction includes the fusion of two gametes, each with a complete haploid ($n$) set of chromosomes. Gametes are formed from specialised diploid ($2n$) cells. The specialised kind of cell division that reduces the chromosome number by half and results in the production of haploid daughter cells is called **meiosis**.

Meiosis occurs during **gametogenesis** in plants and animals, ensuring the production of the haploid phase in the life cycle of sexually reproducing organisms, whereas fertilisation restores the diploid phase.

### 4.1 Key Features of Meiosis

- Meiosis involves **two sequential cycles of nuclear and cell division** called **Meiosis I** and **Meiosis II**, but **only a single cycle of DNA replication**.
- Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the $S$ phase.
- Meiosis involves the **pairing of homologous chromosomes** and **recombination between non-sister chromatids of homologous chromosomes**.
- **Four haploid cells** are formed at the end of Meiosis II.

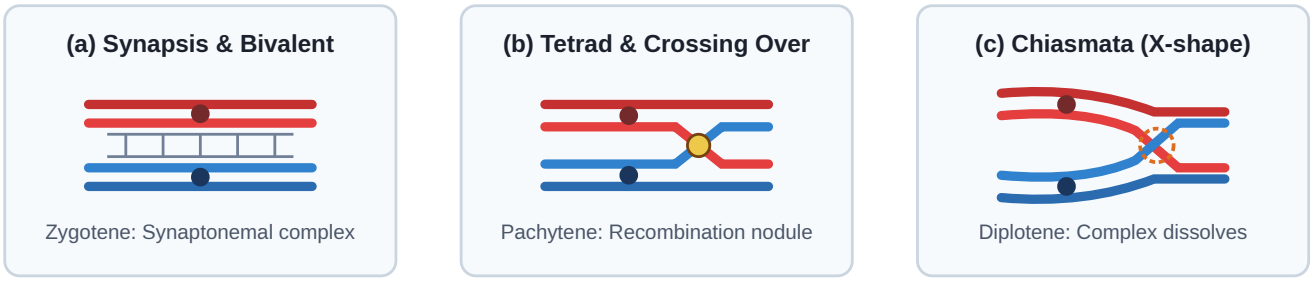
| Meiosis I (Reductional Division) | Meiosis II (Equational Division — Resembles Mitosis) |
|---|---|
| **Prophase I** (Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis) | **Prophase II** |
| **Metaphase I** | **Metaphase II** |
| **Anaphase I** | **Anaphase II** |
| **Telophase I** | **Telophase II** |

### 4.2 Detailed Events of Meiosis I

#### Prophase I

Prophase of the first meiotic division is typically **longer and more complex** when compared to the prophase of mitosis. It is subdivided into five phases based on chromosomal behaviour:

1. **Leptotene:**
   - Chromosomes become gradually visible under the light microscope.
   - The **compaction of chromosomes continues throughout leptotene**.
2. **Zygotene:**
   - Chromosomes start pairing together; this process of association is called **synapsis**.
   - Such paired chromosomes are called **homologous chromosomes**.
   - Electron micrographs of this stage indicate that chromosome synapsis is accompanied by the formation of a complex structure called the **synaptonemal complex**.
   - The complex formed by a pair of synapsed homologous chromosomes is called a **bivalent** or a **tetrad**. However, these are more clearly visible at the next stage.
   - *Note on duration:* The first two stages of Prophase I (leptotene and zygotene) are relatively short-lived compared to the next stage, pachytene.
3. **Pachytene:**
   - During this stage, the **four chromatids of each bivalent chromosome become distinct and clearly appear as tetrads**.
   - Characterised by the appearance of **recombination nodules**, the sites at which **crossing over** occurs between **non-sister chromatids of the homologous chromosomes**.
   - **Crossing over** is the exchange of genetic material between two homologous chromosomes. It is an enzyme-mediated process, and the enzyme involved is called **recombinase**.
   - Crossing over leads to the recombination of genetic material on the two chromosomes. **Recombination between homologous chromosomes is completed by the end of pachytene**, leaving the chromosomes linked at the sites of crossing over.
4. **Diplotene:**
   - Recognised by the **dissolution of the synaptonemal complex** and the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the sites of crossovers.
   - These X-shaped structures are called **chiasmata**.
   - In **oocytes of some vertebrates, diplotene can last for months or years**.
5. **Diakinesis:**
   - The final stage of meiotic Prophase I, marked by the **terminalisation of chiasmata**.
   - Chromosomes are **fully condensed**, and the **meiotic spindle is assembled** to prepare the homologous chromosomes for separation.
   - By the end of diakinesis, the **nucleolus disappears** and the **nuclear envelope also breaks down**.
   - Diakinesis represents the **transition to metaphase**.




**Tutorial Diagram 3: Chromosomal events of Prophase I illustrating (a) Synapsis and bivalent formation with the synaptonemal complex in Zygotene, (b) Four-chromatid tetrad and recombination nodule in Pachytene, and (c) X-shaped Chiasmata in Diplotene.**

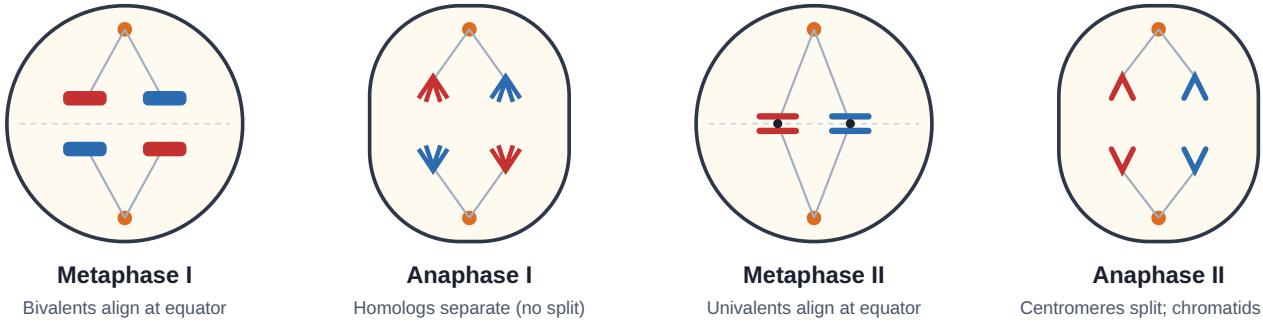
### Metaphase I, Anaphase I, Telophase I, and Interkinesis

- **Metaphase I:** The **bivalent chromosomes align on the equatorial plate** (forming a double metaphasic plate). Microtubules from the opposite poles of the spindle attach to the kinetochore of homologous chromosomes.
- **Anaphase I:** The **homologous chromosomes separate**, while **sister chromatids remain associated at their centromeres** (no splitting of centromeres occurs in Anaphase I). Each pole receives half the chromosome number of the parent cell.
- **Telophase I:** The nuclear membrane and nucleolus reappear, cytokinesis follows, and this two-cell stage is called a **dyad of cells**. Although in many cases the chromosomes do undergo some dispersion, they do not reach the extremely extended state of the interphase nucleus.
- **Interkinesis:** The stage between the two meiotic divisions is called **interkinesis** and is generally short-lived. **There is no replication of DNA during interkinesis.** Interkinesis is followed by Prophase II, a much simpler prophase than Prophase I.

## 4.3 Detailed Events of Meiosis II

In contrast to Meiosis I, **Meiosis II resembles a normal mitosis**:

- **Prophase II:** Meiosis II is initiated immediately after cytokinesis, usually before the chromosomes have fully elongated. The nuclear membrane disappears by the end of Prophase II, and the chromosomes again become compact.
- **Metaphase II:** The chromosomes align at the equator, and the microtubules from opposite poles of the spindle get attached to the **kinetochores of sister chromatids**.
- **Anaphase II:** Begins with the **simultaneous splitting of the centromere** of each chromosome (which was holding the sister chromatids together), allowing the daughter chromosomes to move toward opposite poles of the cell by shortening of microtubules attached to kinetochores.
- **Telophase II:** Meiosis ends with Telophase II, in which the two groups of chromosomes once again get enclosed by a nuclear envelope; cytokinesis follows, resulting in the formation of a **tetrad of cells, i.e., four haploid daughter cells**.




**Tutorial Diagram 4: Key comparative stages of Meiosis I and Meiosis II (for a parent cell with 2n = 4) showing bivalent alignment and homolog separation in Meiosis I versus univalent alignment and centromere splitting in Meiosis II.**

## 4.4 Significance of Meiosis

- **Conservation of Chromosome Number:** Meiosis is the mechanism by which **conservation of specific chromosome number of each species is achieved across generations in sexually reproducing organisms**, even though the process, *per se*,

paradoxically results in the reduction of chromosome number by half.

- **Genetic Variability and Evolution:** Meiosis **increases the genetic variability** in the population of organisms from one generation to the next (primarily via crossing over during pachytene). Variations are very important for the process of **evolution**.

## 5. Quantitative Analysis & Master Comparisons

A frequent NEET UG question pattern tests how the **number of chromosomes ($N$ or $2n$)** and the **amount of DNA content ($C$)** change through the cell cycle, mitosis, and meiosis.

### 5.1 Tracking Chromosome Number and DNA Content

Consider a diploid parent cell starting in $\mathrm{G}_1$ phase with chromosome number $2n$ and DNA content $2C$:

| Stage of Cell Cycle / Division | Chromosome Number per Cell | DNA Content per Cell | Key Biological Reason |
|---|---|---|---|
| **$\mathrm{G}_1$ Phase** | $2n$ | $2C$ | Each chromosome has 1 chromatid (unreplicated DNA). |
| **After $\mathrm{S}$ Phase / $\mathrm{G}_2$ Phase** | $2n$ | $4C$ | DNA replicates ($2C \rightarrow 4C$), forming 2 sister chromatids held by 1 centromere; chromosome number stays $2n$. |
| **Mitotic Prophase & Metaphase** | $2n$ | $4C$ | Condensed chromosomes each have 2 sister chromatids. |
| **Mitotic Anaphase & Telophase (within whole dividing cell prior to cytokinesis)** | $4n$ (transiently in cell; $2n$ per pole) | $4C$ ($2C$ per daughter nucleus) | Centromeres split; each chromatid becomes an independent daughter chromosome moving to a pole. |
| **After Mitotic Cytokinesis (per daughter cell)** | $2n$ | $2C$ | Cytoplasm divides; baseline diploid state restored. |
| **Meiosis I: Prophase I & Metaphase I** | $2n$ (paired as $n$ bivalents) | $4C$ | Homologous chromosomes pair into bivalents/tetrads. |
| **After Meiosis I / Telophase I / Interkinesis (per dyad cell)** | $n$ | $2C$ | Homologous chromosomes separate into 2 cells; each chromosome still has 2 chromatids. |
| **After Meiosis II / Telophase II (per haploid gamete/tetrad cell)** | $n$ | $1C$ | Centromeres split in Anaphase II; sister chromatids separate into 4 haploid cells. |

### 5.2 Important Quantitative Formulas for NEET UG

- $\text{Number of Bivalents (or Tetrads) in Prophase I / Metaphase I} = n = \dfrac{\text{Diploid chromosome number } (2n)}{2}$
- $\text{Total number of chromatids in Prophase / Metaphase (Mitosis or Meiosis I)} = 2 \times (2n) = 4n$
- $\text{Number of mitotic divisions required to produce } X \text{ cells from 1 cell (asynchronous)} = X - 1$
- $\text{Number of generations (cycles) of mitosis to produce } X \text{ cells from 1 cell} = k \text{ where } 2^k = X$
- $\text{Number of cells produced from 1 cell after } k \text{ mitotic cycles} = 2^k$
- $\text{Number of haploid cells (gametes/spores) produced from 1 diploid meiocyte after complete meiosis} = 4$

### 5.3 Key Conceptual Distinctions & Special Cases

| Comparison / Question | Scientific Explanation |
|---|---|
| **Anaphase of Mitosis vs. Anaphase I of Meiosis** | In **Anaphase of Mitosis**, the centromere of each chromosome splits and sister chromatids separate to opposite poles. In **Anaphase I of Meiosis**, centromeres do *not* split; homologous chromosomes separate to opposite poles while sister chromatids remain associated at their centromeres. |
| **Equal vs. Unequal Daughter Cells in Meiosis** | **Equal-sized four daughter cells:** Spermatogenesis in animals (forming four equal spermatids) and microsporogenesis in flowering plants (forming four equal microspores).<br>**Unequal-sized daughter cells:** Oogenesis in animals (forming one large ovum and tiny polar bodies) and megasporogenesis in gymnosperms/angiosperms. |
| **Haploid cells in higher plants where cell division does NOT occur** | Synergids, antipodal cells, and egg cell inside the mature female gametophyte (embryo sac), or male gametes after pollen maturation. |
| **Can there be mitosis without DNA replication in $\mathrm{S}$ phase?** | **No.** Without DNA replication in the $\mathrm{S}$ phase, daughter cells would not receive intact, duplicated genomes; genetic checkpoints prevent entry into mitosis until DNA synthesis is complete. |
| **Can there be DNA replication without cell division?** | **Yes.** DNA replication in the $\mathrm{S}$ phase creates $4C$ DNA prior to $\mathrm{M}$ phase, and in certain specialized cases (such as polyploidy/endoreduplication or syncytial stages where cytokinesis fails), DNA replicates without immediate cell division. |

## 6. Worked Examples

### Worked Example 1: Onion Root Tip Chromosome & DNA Calculation

**Problem:** An onion root tip cell undergoing mitosis has $16$ chromosomes in each cell ($2n = 16$). If the DNA content of a daughter cell immediately after $\mathrm{M}$ phase is $2C$:
(a) How many chromosomes will the cell have at $\mathrm{G}_1$ phase, after $\mathrm{S}$ phase, and after $\mathrm{M}$ phase?
(b) What will be the DNA content of the cell at $\mathrm{G}_1$, after $\mathrm{S}$, and at $\mathrm{G}_2$?

**Solution:**

- **(a) Chromosome Number:** In $\mathrm{G}_1$ phase, the cell is diploid with $2n = 16$ chromosomes. During $\mathrm{S}$ phase, DNA replicates, converting each chromosome from a $1\text{-chromatid}$ structure to a $2\text{-chromatid}$ structure held by a single centromere; thus, the chromosome number does *not* increase. After $\mathrm{M}$ phase (mitosis, an equational division), each daughter cell receives a full set of chromosomes.
$$\text{Chromosomes at } \mathrm{G}_1 = 16, \quad \text{After } \mathrm{S} = 16, \quad \text{After } \mathrm{M} = 16$$
- **(b) DNA Content:** Immediately after $\mathrm{M}$ phase, the cell enters $\mathrm{G}_1$ phase with $2C$ DNA. During the $\mathrm{S}$ phase, the amount of DNA per cell doubles ($2C \rightarrow 4C$) and remains at $4C$ throughout the $\mathrm{G}_2$ phase until the cell divides in $\mathrm{M}$ phase.
$$\text{DNA at } \mathrm{G}_1 = 2C, \quad \text{After } \mathrm{S} = 4C, \quad \text{At } \mathrm{G}_2 = 4C$$

### Worked Example 2: Bivalents, Chromatids, and Gamete DNA in Meiosis

**Problem:** A diploid meiocyte (germ mother cell) has $2n = 24$ chromosomes and $40\ \text{pg}$ (picograms) of DNA at the $\text{G}_1$ phase. Determine:
(a) Number of bivalents formed during Zygotene/Pachytene.
(b) Number of chromosomes and amount of DNA in each cell at Telophase I (dyad stage).
(c) Number of chromosomes and amount of DNA in each haploid gamete after Telophase II.

**Solution:**

- **At $\text{G}_1$ phase:** Chromosome number $2n = 24$; DNA content $2C = 40\ \text{pg}$.
- **After $\text{S}$ phase (entering Prophase I):** Chromosome number $= 24$; DNA content $4C = 80\ \text{pg}$.
- **(a) Number of bivalents in Zygotene/Pachytene:** Each bivalent consists of a pair of synapsed homologous chromosomes.

$$\text{Number of bivalents} = n = \frac{24}{2} = 12\ \text{bivalents}$$

- **(b) At Telophase I (per dyad cell):** Anaphase I separates homologous chromosomes into two cells without splitting centromeres.

$$\text{Chromosomes per cell} = n = 12, \quad \text{DNA content per cell} = 2C = \frac{80\ \text{pg}}{2} = 40\ \text{pg}$$

- **(c) After Telophase II (per gamete):** Anaphase II splits the centromeres and separates sister chromatids into four haploid cells.

$$\text{Chromosomes per gamete} = n = 12, \quad \text{DNA content per gamete} = 1C = \frac{40\ \text{pg}}{2} = 20\ \text{pg}$$

## 7. NEET UG Practice Questions with Worked Solutions

*Note: The following are original, exam-aligned practice questions designed according to the NEET UG examination standard.*

#### Practice Question 1 (Single-Correct MCQ)

In a typical human cell in culture with a $24\text{-hour}$ cell cycle duration, what is the approximate duration of the $\text{M}$ phase and Interphase, respectively?

(1) $12\ \text{hours}$ and $12\ \text{hours}$

(2) About $1\ \text{hour}$ and more than $95\%$ of the cell cycle duration

(3) $90\ \text{minutes}$ and $22.5\ \text{hours}$

(4) Less than $5\ \text{minutes}$ and $99.5\%$ of the cell cycle duration

**Correct Answer: (2)**
**Worked Solution:** Human cells in culture divide once in approximately every $24\ \text{hours}$. In this $24\text{-hour}$ average duration, cell division proper ($\text{M}$ phase) lasts for only about $1\ \text{hour}$, whereas Interphase lasts more than $95\%$ of the total duration of the cell cycle. (Note: $90\ \text{minutes}$ is the total cell cycle duration of yeast).

**Practice Question 2 (Single-Correct MCQ)**

During the $\mathrm{S}$ phase of interphase in an animal cell, which of the following events occur simultaneously in the nucleus and cytoplasm, respectively?

(1) Chromosome condensation in the nucleus and spindle fibre assembly in the cytoplasm

(2) DNA replication in the nucleus and centriole duplication in the cytoplasm

(3) Centriole duplication in the nucleus and DNA replication in the cytoplasm

(4) Splitting of centromeres in the nucleus and furrow formation in the cytoplasm

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**Correct Answer: (2)**
**Worked Solution:** In animal cells, during the $\mathrm{S}$ (synthesis) phase of interphase, DNA replication begins inside the nucleus while the centriole (centrosome) duplicates in the cytoplasm.

**Practice Question 3 (Single-Correct MCQ)**

Which of the following accurately describes cells in the quiescent stage ($\mathrm{G}_0$) of the cell cycle, such as adult mammalian heart cells?

(1) They exit the $\mathrm{G}_2$ phase, become metabolically inactive, and undergo programmed cell death.

(2) They exit the $\mathrm{G}_1$ phase, remain metabolically active, and do not proliferate unless called on to do so.

(3) They are arrested at Metaphase and continuously synthesise DNA without dividing.

(4) They exit the $\mathrm{S}$ phase with $4C$ DNA content and lose all cellular organelles.

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**Correct Answer: (2)**
**Worked Solution:** Cells that do not divide further exit the $\mathrm{G}_1$ phase to enter an inactive stage called the quiescent stage ($\mathrm{G}_0$) of the cell cycle. Cells in this stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism (such as replacing cells lost due to injury).

**Practice Question 4 (Single-Correct MCQ)**

A somatic root tip cell of a plant species has $2n = 14$ chromosomes at the $\mathrm{G}_1$ phase. What will be the number of chromosomes and the number of chromatids present in this cell at the $\mathrm{G}_2$ phase?

(1) $14$ chromosomes and $14$ chromatids

(2) $28$ chromosomes and $28$ chromatids

(3) $14$ chromosomes and $28$ chromatids

(4) $7$ chromosomes and $14$ chromatids

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**Correct Answer: (3)**
**Worked Solution:** During the $\mathrm{S}$ phase (prior to $\mathrm{G}_2$), DNA content doubles ($2C \rightarrow 4C$), so each chromosome duplicates to consist of $2$ sister chromatids attached at a single centromere. However, the chromosome number remains unchanged ( $2n = 14$). Therefore, at $\mathrm{G}_2$, there are $14$ chromosomes and $14 \times 2 = 28$ chromatids.

**Practice Question 5 (Single-Correct MCQ)**

If a cytogeneticist wants to study the morphology of chromosomes most clearly under a light microscope, which stage of mitosis should be selected, and what event marks the start of this stage?

(1) Prophase; marked by the initiation of chromatin condensation

(2) Metaphase; marked by the complete disintegration of the nuclear envelope

(3) Anaphase; marked by the splitting of centromeres

(4) Telophase; marked by the clustering of chromosomes at opposite spindle poles

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**Correct Answer: (2)**
**Worked Solution:** The complete disintegration of the nuclear envelope marks the start of the second phase of mitosis, Metaphase. By this stage, condensation of chromosomes is completed and they can be observed clearly under the microscope; hence, Metaphase is the stage at which the morphology of chromosomes is most easily studied.

**Practice Question 6 (Single-Correct MCQ)**

During mitotic Anaphase, as each daughter chromosome moves away from the equatorial plate towards the opposite pole:

(1) The arms of the chromosome lead, while the centromere trails behind.

(2) The centromere remains directed towards the pole at the leading edge, with the chromosome arms trailing behind.

(3) The homologous chromosomes separate without splitting of the centromere.

(4) Spindle fibres attach directly to the telomeres of the chromosomes.

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**Correct Answer: (2)**
**Worked Solution:** At the onset of Anaphase, each chromosome arranged at the metaphase plate splits simultaneously at the centromere. As each daughter chromosome migrates away from the equatorial plate, the centromere remains directed towards the pole and hence is at the leading edge, with the arms of the chromosome trailing behind.

### Practice Question 7 (Matching-Type MCQ)

Match the substages of Prophase I in Column I with their characteristic cytological events in Column II:

| Column I (Substage of Prophase I) | Column II (Characteristic Event) |
|---|---|
| A. Zygotene | (i) Appearance of recombination nodules and crossing over by recombinase |
| B. Pachytene | (ii) Dissolution of synaptonemal complex and appearance of X-shaped chiasmata |
| C. Diplotene | (iii) Terminalisation of chiasmata and assembly of meiotic spindle |
| D. Diakinesis | (iv) Synapsis of homologous chromosomes and synaptonemal complex formation |

(1) A-(iv), B-(i), C-(ii), D-(iii)

(2) A-(i), B-(iv), C-(iii), D-(ii)

(3) A-(iv), B-(ii), C-(i), D-(iii)

(4) A-(ii), B-(i), C-(iv), D-(iii)

**Correct Answer: (1)**

**Worked Solution:**

- **Zygotene (A-iv):** Pairing of homologous chromosomes (synapsis) and formation of the synaptonemal complex.
- **Pachytene (B-i):** Four chromatids of each bivalent become distinct as tetrads; recombination nodules appear and recombinase-mediated crossing over occurs.
- **Diplotene (C-ii):** Dissolution of the synaptonemal complex and formation of X-shaped chiasmata.
- **Diakinesis (D-iii):** Terminalisation of chiasmata, full chromosome condensation, and assembly of the meiotic spindle.

### Practice Question 8 (Statement-Based MCQ)

Read the following two statements carefully and choose the correct option:

**Statement I:** Recombination between homologous chromosomes is initiated in Zygotene and completed by the end of Diplotene.

**Statement II:** Crossing over is an enzyme-mediated process involving the enzyme recombinase between non-sister chromatids of homologous chromosomes.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct, but Statement II is incorrect.

(4) Statement I is incorrect, but Statement II is correct.

**Correct Answer: (4)**

**Worked Solution:** Statement I is incorrect because crossing over occurs during **Pachytene** (at recombination nodules) and **recombination between homologous chromosomes is completed by the end of Pachytene**, leaving the chromosomes linked at the sites of crossing over. Statement II is completely accurate.

### Practice Question 9 (Assertion-Reason MCQ)

**Assertion (A):** Plant cells undergo cytokinesis by forming a cell-plate in the centre of the cell that grows outward, rather than by plasma membrane furrow formation.
**Reason (R):** Plant cells are enclosed by a relatively inextensible cell wall, and the cell-plate formed during cytokinesis represents the middle lamella between the walls of two adjacent cells.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

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**Correct Answer: (1)**
**Worked Solution:** Because plant cells are enclosed by a relatively inextensible cell wall, they cannot pinch inward via a peripheral plasma membrane furrow like animal cells. Instead, wall formation starts in the centre of the cell with a precursor called the cell-plate (which represents the middle lamella between adjacent cell walls) and grows outward to meet the existing lateral walls. Thus, both (A) and (R) are true, and (R) explains (A).

### Practice Question 10 (Assertion-Reason MCQ)

**Assertion (A):** In Anaphase I of meiosis, the chromosome number at each spindle pole is reduced to half of the parental diploid cell, yet the DNA content at each pole remains $2C$.
**Reason (R):** During Anaphase I, homologous chromosomes separate towards opposite poles while sister chromatids remain associated at their centromeres.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

---

**Correct Answer: (1)**
**Worked Solution:** At Metaphase I, a diploid cell has $2n$ chromosomes and $4C$ DNA. In Anaphase I, homologous chromosomes separate to opposite poles without centromere splitting, so each pole receives $n$ chromosomes. Since each of these $n$ chromosomes still consists of two sister chromatids associated at the centromere, the DNA content per pole (and per resulting dyad cell) is half of $4C$, which is $2C$. Hence, both (A) and (R) are true and (R) is the exact explanation of (A).

**Practice Question 11 (Multi-Statement Evaluation MCQ)**

Which of the following statements regarding **Interkinesis** are correct?

I. It is the stage between Meiosis I and Meiosis II and is generally short-lived.

II. A complete round of DNA replication takes place during interkinesis to restore $4C$ DNA content.

III. During Telophase I and interkinesis, chromosomes may undergo some dispersion, but they do not reach the extremely extended state of the interphase nucleus.

IV. Interkinesis is followed directly by Prophase II, which is much simpler than Prophase I.

(1) I, II, and IV only

(2) I, III, and IV only

(3) II and III only

(4) I, II, III, and IV

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**Correct Answer: (2)**

**Worked Solution:** Statement II is false because **there is no replication of DNA during interkinesis**. Meiosis involves two sequential cycles of nuclear and cell division (Meiosis I and II) but only a single cycle of DNA replication (which occurs prior to Meiosis I). Statements I, III, and IV are completely correct.

**Practice Question 12 (Quantitative MCQ)**

A pollen grain (haploid microspore) of a flowering plant contains $12$ chromosomes and $15\ \text{pg}$ of DNA. What would have been the number of bivalents and the total amount of DNA in its microspore mother cell (meiocyte) at the **Pachytene** stage of Meiosis I?

(1) $12$ bivalents and $30\ \text{pg}$ of DNA

(2) $24$ bivalents and $60\ \text{pg}$ of DNA

(3) $12$ bivalents and $60\ \text{pg}$ of DNA

(4) $6$ bivalents and $30\ \text{pg}$ of DNA

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**Correct Answer: (3)**

**Worked Solution:**

- A haploid product of meiosis (pollen grain/microspore) has $n$ chromosomes and $1C$ DNA. Thus, $n = 12$ and $1C = 15\ \text{pg}$.
- The diploid microspore mother cell (meiocyte) has $2n = 24$ chromosomes. At Pachytene, homologous chromosomes pair to form $n$ bivalents, so there are $12$ bivalents.
- At Pachytene (which follows the $\text{S}$ phase), the DNA content of the meiocyte is $4C$:

$$4C = 4 \times 15\ \text{pg} = 60\ \text{pg}$$

**Practice Question 13 (Conceptual MCQ)**

Identify the correct chronological order of the following events during a meiotic division:

A. Dissolution of the synaptonemal complex
B. Synapsis of homologous chromosomes
C. Splitting of centromeres of sister chromatids
D. Crossing over mediated by recombinase
E. Terminalisation of chiasmata

(1) B → D → A → E → C

(2) B → A → D → E → C

(3) D → B → A → E → C

(4) B → D → E → A → C

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**Correct Answer: (1)**
**Worked Solution:**

1. **B:** Synapsis of homologous chromosomes occurs in **Zygotene**.
2. **D:** Crossing over mediated by recombinase occurs in **Pachytene**.
3. **A:** Dissolution of the synaptonemal complex marks the beginning of **Diplotene**.
4. **E:** Terminalisation of chiasmata marks **Diakinesis**.
5. **C:** Splitting of centromeres of sister chromatids occurs much later in **Anaphase II**.

Therefore, the sequence is B → D → A → E → C.

**Practice Question 14 (Single-Correct MCQ)**

In oocytes of some vertebrates, which substage of Meiotic Prophase I can remain arrested and last for months or years?

(1) Leptotene

(2) Zygotene

(3) Pachytene

(4) Diplotene

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**Correct Answer: (4)**
**Worked Solution:** In oocytes of some vertebrates, **diplotene** (characterised by dissolution of the synaptonemal complex and X-shaped chiasmata) can last for months or years.

## 8. Common Pitfalls & Rapid Revision Checklist

### High-Frequency NEET UG Traps to Avoid

- **Trap 1 — Chromosome vs. DNA doubling in $\mathrm{S}$ phase:** Students often assume both chromosome number and DNA content double in the $\mathrm{S}$ phase. *Correction:* Only DNA content doubles ($2C \rightarrow 4C$); chromosome number remains $2n$.
- **Trap 2 — Bivalent visibility vs. formation:** A bivalent/tetrad is *formed* during Zygotene due to synapsis, but the four chromatids of each bivalent become *distinct and clearly visible* as a tetrad only in **Pachytene**.
- **Trap 3 — When recombination finishes:** Recombination is completed by the **end of Pachytene** (not Diplotene or Diakinesis). At Diplotene, the chromosomes are already recombined and stay linked at chiasmata.
- **Trap 4 — Centromere splitting in Meiosis:** Centromeres **do NOT split in Anaphase I**; they split only in **Anaphase II** (and in Mitotic Anaphase).
- **Trap 5 — Kinetochore vs. Centromere:** The centromere is the primary constriction holding sister chromatids together, whereas **kinetochores** are the small disc-shaped structures on the surface of the centromere where spindle fibres attach.

## 9. Concise Final Recap

- **Cell Cycle Phases:** Interphase ($\mathrm{G_1} \rightarrow \mathrm{S} \rightarrow \mathrm{G_2}$, taking $> 95\%$ of human $24\,\mathrm{h}$ cycle) + $\mathrm{M}$ Phase ($\approx 1\,\mathrm{h}$). Yeast cell cycle $= 90\,\mathrm{min}$.
- **Mitosis (Equational Division):**
  - *Prophase:* Chromatin condenses; centrosomes move to poles radiating asters (2 asters + spindle fibres = mitotic apparatus); ER, Golgi, nucleolus, and nuclear envelope disappear by the end.
  - *Metaphase:* Nuclear envelope completely disintegrates; chromosomes maximally condensed; spindle fibres attach to kinetochores; chromosomes align at the metaphase plate.
  - *Anaphase:* Centromeres split; daughter chromosomes move to poles with centromeres leading and arms trailing.
  - *Telophase:* Chromosomes decondense at poles; nuclear envelope, nucleolus, ER, and Golgi complex reform.
  - *Cytokinesis:* Furrow in animals; centrifugal cell-plate (precursor to middle lamella) in plants; failure of cytokinesis after karyokinesis yields a multinucleate syncytium (e.g., liquid endosperm in coconut).
- **Meiosis (Reductional Division):** Two nuclear divisions, one DNA replication ($\mathrm{S}$ phase before Meiosis I), yielding $4$ haploid ($n, 1C$) daughter cells.
  - *Prophase I:* Leptotene (compaction) $\rightarrow$ Zygotene (synapsis, synaptonemal complex, bivalent) $\rightarrow$ Pachytene (distinct tetrad, recombination nodules, crossing over via recombinase completed) $\rightarrow$ Diplotene (dissolution of synaptonemal complex, chiasmata, prolonged in vertebrate oocytes) $\rightarrow$ Diakinesis (terminalisation of chiasmata, spindle assembly, disappearance of nucleolus and nuclear envelope).
  - *Metaphase I & Anaphase I:* Bivalents align at equator; homologous chromosomes separate (sister chromatids remain attached). Telophase I forms a dyad of cells ($n, 2C$) followed by Interkinesis (no DNA replication).
  - *Meiosis II:* Resembles mitosis; centromeres split in Anaphase II; Telophase II forms a tetrad of four haploid cells ( $n, 1C$).